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Series arc fault detection via arcing physical characteristics-guided feature extraction: a dual-signal multi-timescale method

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ABSTRACT

Series arc faults (SAFs) are a primary cause of electrical fires in building distribution systems, requiring reliable detection to mitigate fire risks and ensure operational safety. Existing methods, though effective under simplified scenarios, suffer from false alarms and missed detections in complex scenarios involving diverse load types, circuit topologies, arc-generating modes, and arc-like conditions. To address the issues, this paper proposes a SAF detection method that integrates a dual-signal multi-timescale feature extraction framework, an arc fault decision tree, and arc fault decision criteria. Firstly, an experimental platform simulating SAFs in building distribution systems is established, and the arcing characteristics are analyzed through key electrical parameters—arc voltage, current, and zero-sequence current coupling signal—across four stages: arc-igniting, arcing, arc-extinguishing, and zero-current under different arc-generating modes. Secondly, based on these characteristics, with a particular focus on the current and zero-sequence current coupling signal, a comprehensive arc fault feature set is constructed by extracting features from both single-cycle and multi-cycle perspectives. Thirdly, the arc fault decision tree is trained using the constructed feature set, and the arc fault decision criteria are designed based on the persistence characteristics of arc faults. Finally, the superiority of the proposed method is demonstrated through offline tests, online verification on a self-developed prototype, anti-interference experiments, and a comparative analysis. Experimental results show that the proposed method achieves a detection accuracy of 99.20 % under test conditions encompassing thirteen loads, four circuit topologies, and two arc-generating modes, outperforming five representative benchmark methods under identical test conditions. Furthermore, during 360 online tests covering both training conditions and previously unseen real-world operating conditions, the proposed method satisfied standard detection time requirements in 354 tests while maintaining zero false alarms under arc-like conditions such as load switching.

1. Introduction

With the growing demand for electricity driven by economic and social development, electrical fire incidents have increased significantly [1]. According to statistics from the National Fire and Rescue Administration, a total of 745000 fire incidents were reported nationwide from January to October 2023, of which 29.1 % were caused by electrical faults [2]. In building distribution systems, common issues such as poor cable contact, conductor fracture, insulation damage, and insulation aging can lead to arc faults [3]. These faults generate temperatures exceeding thousands of degrees Celsius, which can easily ignite nearby combustible materials, resulting in electrical fires [4,5]. Arc faults account for over 50 % of the causes of electrical fires [6].

Based on the relationship between fault location and load location, arc faults are classified into parallel arc faults (PAFs) and series arc faults (SAFs). When PAFs occur, the loop current significantly increases, enabling traditional protective devices (such as miniature circuit breakers) to detect the fault and open the faulty circuit promptly. In contrast, SAFs behave like a time-varying load, resulting in a lower loop current than the load current during normal operation, which makes timely detection by traditional protective devices challenging [7]. To prevent electrical fires caused by SAFs, various detection methods have been proposed by researchers worldwide, some of which have been implemented in embedded systems and commercialized. However, evaluations of representative market-available products reveal that they provide effective protection only in simple scenarios involving few loads, a single circuit topology, and one arc-generating mode. These products exhibit significant rates of false alarms and missed detections in

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more complex scenarios characterized by diverse loads, multiple circuit topologies, two arc-generating modes, and various arc-like operating conditions [8]. Consequently, developing more robust SAF detection methods is essential for advancing arc fault detection devices (AFDDs) and reducing electrical fire risks.

SAF detection methods universally employ a two-stage framework: (1) feature extraction through expert-designed or deep learning-based approaches to capture arc fault signatures, followed by (2) state discrimination using physics-constrained thresholds or data-driven decision boundaries to classify operational conditions. Based on the strategies adopted for feature extraction and state discrimination, existing methods can be broadly categorized into two groups: those utilizing time–frequency domain features with machine learning, and those employing deep learning with end-to-end detection.

During a SAF, distinct distortions manifest in electrical parameters. This is most evident in the current waveform, which exhibits signatures including amplitude attenuation, flat-shoulder patterns, high-frequency spikes, ripple noise, and asymmetric waveforms [9-11]. These distortions, along with complementary variations in voltage and zero-sequence current coupling signal (ZSCCS) waveforms, provide a comprehensive basis for SAF detection [12-14]. Reference [15] proposes a SAF detection method utilizing sym4 wavelet decomposition for current signal processing, where statistical features (integral, variance, kurtosis, Shannon entropy) are classified through XGBoost. Reference [16] introduces an L1/L2 norm-based metric to quantify arcing current fluctuations, integrating time–frequency indicators (kurtosis, shape factor, crest factor, spectral centroid) into a grid search-optimized random forest classifier. Reference [17] fuses time-domain (integral, Shannon entropy) and frequency-domain (harmonic amplitudes) features, applying empirical thresholds for robust SAF detection. In [18], time–frequency features (fundamental component, harmonic components, zero-current duration, entropy) undergo sensitivity analysis before time-variance-accuracy-optimized random forest classification. Reference [19] implements chirp zeta transform-derived low-frequency spectral features (inter-window spectral means, frequency band ratios) with dynamically adjusted thresholds for operational state discrimination. The aforementioned methods quantify specific current waveform distortions induced by SAFs, enabling reliable detection in simple scenarios. However, their performance significantly deteriorates in complex scenarios. This limitation stems from the fact that these methods fail to account for real-world complexities, including diverse loads, complex circuit topologies, various arc-like operating conditions, and different arc-fault causes. Current waveform characteristics are highly sensitive to load types, and the parallel connection of fault branches with normal branches further modifies these characteristics [20]. Consequently, these methods lack the discrimination capability to distinguish arc-like conditions or identify fault conditions where faulty and normal branches are connected in parallel, resulting in potential misoperation and detection failures.

To enhance the accuracy of SAF detection in complex scenarios, researchers have refined detection methods by expanding load classification criteria, increasing arc feature indicators, and utilizing alternative detection signals. Reference [21] classifies loads into resistive, inductive, and switching types, extracting statistical features (range, mean, integral, deviation) for k-nearest neighbor-based detection models. Reference [22] employs enhanced ensemble empirical mode decomposition to extract thirty-three arc fault features, reduced to ten optimal dimensions via TreeBagger feature selection for random forest classification. Reference [23] employs Kalman filtering to estimate current steady-states, generating residuals and third-order differences for fuzzy logic threshold adaptation. Reference [24] proposes a neural network-based SAF detection method utilizing multi-band spectral features (mean, deviation, skewness, kurtosis, max) derived from discrete Fourier, chirp zeta, and wavelet transforms. In [25], arc voltage is employed as the detection signal, and a method based on the similarity of the 3rd, 5th, and 7th harmonic parameters of arc voltage is proposed. Despite these advancements, challenges persist in applying these methods to building distribution systems. The core limitation lies in the failure of these methods to account for either the interference from arc-like conditions or the variations in arc characteristics under different arc-generating modes. Specifically, since interferences from arc-like conditions closely resemble the waveform of certain fault conditions, and arc characteristics from poor contact differ from those from insulation aging, these methods cannot extract reliable exclusive features—ultimately leading to high rates of false alarms and missed detections when applied to real-world distribution systems [26].

Deep learning eliminates subjectivity in fault detection through autonomous feature extraction and training, circumventing manual feature engineering and empirical thresholds. Convolutional and recurrent neural networks have shown particular efficacy in SAF detection. For instance, in [27], raw current signals are input into a convolutional neural network for SAF detection, utilizing a four-layer convolutional architecture. Reference [28] employs empirical mode decomposition to extract the two most discriminative intrinsic mode functions, enabling convolutional neural network-based SAF detection. In [29], the current signal is processed by a Wiener filter to reduce noise, and a residual network with a high-frequency feature attention module is trained to detect SAFs. Reference [30] converts current signals into relative position matrices, processing them as two-dimensional images via a hybrid-attention residual network for precise SAF detection. In [31], wavelet transforms convert current signals into phase space images for training a residual network, enabling robust fault detection. Despite their excellent performance in simple scenarios, significant challenges remain in applying these methods to complex scenarios. First, their effectiveness highly depends on training datasets, yet comprehensively covering all operating conditions in building distribution systems is difficult. Second, their substantial computational and storage demands make deployment on resource-constrained embedded systems impractical [32]. Addressing these challenges is essential for enhancing the practical applicability of deep learning-based detection methods.

Table 1 summarizes the testing conditions of existing SAF detection methods, revealing that experimental validations are predominantly conducted in simplified scenarios involving few load types, fixed circuit topologies, and a single arc-generating mode. Such simplified scenarios differ significantly from real-world distribution systems, which incorporate diverse load types, complex circuit topologies, multiple arc-generating modes, and various arc-like operating conditions. The high rates of false alarms and missed detections observed in building distribution systems are primarily attributed to the following factors: (1) insufficient analysis of the differences between arc faults caused by poor contact and those resulting from insulation aging restricts protection coverage to only specific fault types; (2) failure to comprehensively quantify distinctive characteristics of SAFs based on their physical properties results in performance degradation under untrained operating conditions; (3) excessive focus on accuracy metrics overlooks critical practical considerations such as model complexity, real-time performance, and interpretability, substantially diminishing their potential for engineering applications.

To overcome these limitations, this paper proposes a SAF detection method. Its key innovation is an arcing physical characteristics-guided feature extraction framework, which fuses two signal sources and extracts features from both single and multiple time-window dimensions. First, an experimental platform is established to simulate SAFs in building distribution systems. Second, a comparative analysis of poor contact-induced and insulation aging-induced arc faults is conducted across four defined stages—arc-igniting, arcing, arc-extinguishing, and zero-current—revealing stage-specific electrical waveform differences attributable to distinct arc-generating modes. Third, based on the physical characteristics of arc faults under different arc-generating modes, features are extracted from both current and ZSCCS across single-cycle and multi-cycle dimensions. Fourth, leveraging the simplicity and interpretability of decision trees, preliminary circuit state

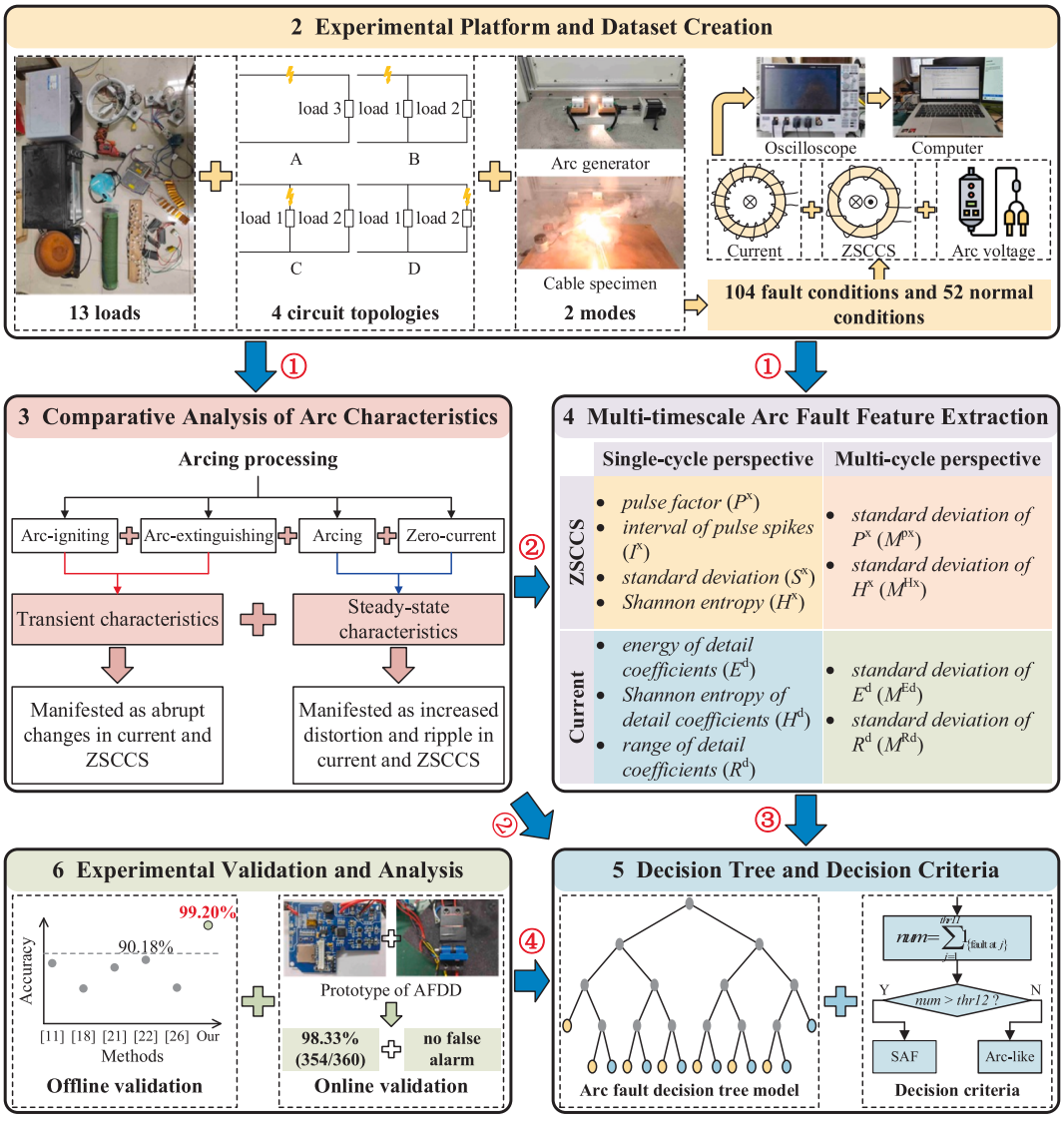
Table 1
Testing conditions of existing SAF detection methods.

References	NL	NCT	NAGM	NOC	OVM (yes/no)	References	NL	NCT	NAGM	NOC	OVM (yes/no)
[15]	4	1	1	4	no	[17]	8	1	2	16	yes
[28]	5	1	1	5	no	[25]	4	4	1	16	yes
[29]	5	1	1	5	no	[27]	8	1	2	16	yes
[30]	5	1	1	5	no	[10]	6	4	1	24	no
[31]	6	1	1	6	no	[12]	6	4	1	24	no
[22]	7	1	1	7	no	[11]	8	4	1	32	yes
[16]	8	1	1	8	no	[14]	10	4	1	40	yes
[18]	8	1	1	8	no	[23]	10	4	1	40	no
[19]	9	1	1	9	no	[24]	10	4	1	40	no
[9]	12	1	1	12	yes	[26]	12	4	1	48	no
[21]	14	1	1	14	no	**Ours**	**13**	**4**	**2**	**104**	**yes**

NL: number of loads; NCT: number of circuit topologies; NAGM: number of arc-generating modes; NOC: number of operating conditions; OVM: online verification on microcontrollers.

identification is achieved using the constructed feature set. Fifth, by exploiting the fundamental difference in persistence between arc-like conditions and SAFs, the arc fault decision criteria are designed to realize final state identification. Finally, comprehensive validations and comparative analyses are conducted, including online validation via a self-developed AFDD prototype. The main contributions of this paper are as follows:

(1) Reveals the distinct physical characteristics of two arc types through a novel four-stage comparative analysis. The arcing



①: provide data support; ②: provide physical corroboration; ③: provide feature basis; ④: provide experimental validation

Fig. 1. Content architecture and inter-section logic of this paper.

process is divided into four stages: arc-igniting, arcing, arc-extinguishing, and zero-current. Quantitative features extracted from arc voltage, current, and ZSCCS demonstrate that poor-contact arcs exhibit stronger transient pulses, while insulation-aging arcs display more prominent high-frequency fluctuations and stochastic behavior. These findings provide a physical foundation for constructing a discriminative feature set to reliably detect SAFs from both origins.

(2) Proposes a dual-signal multi-timescale feature extraction framework for SAF detection. Specifically, seven single-cycle features are extracted from the current and ZSCCS to quantify the transient pulse, high-frequency fluctuation, and stochastic properties of arc faults. Furthermore, to explicitly capture the randomness over a longer timescale, four of these features are selected, and their standard deviations across multiple consecutive cycles are computed to form a set of multi-cycle features. Validated under a comprehensive set of conditions encompassing thirteen loads, four circuit topologies, and two arc-generating modes, the constructed feature set achieves a detection accuracy of 99.20 % (9285/9360), substantially outperforming five state-of-the-art benchmark methods under identical test conditions.

(3) Designs a two-stage decision logic employing a decision tree followed by definitive criteria. The decision tree is first trained on the constructed feature set to provide a preliminary state assessment. The arc fault decision criteria, leveraging the persistence characteristic of arc faults, then discriminate genuine faults from arc-like interference. Extensive validation on a self-developed AFDD, covering training conditions and unseen real-world conditions, demonstrates a 98.33 % (354/360) pass rate for detection timeliness and zero false alarms under arc-like conditions such as load switching.

Fig. 1 illustrates the overall structure and section logic of this paper. The research follows a systematic workflow, beginning with the development of the experimental platform (Section 2), which provides the foundation for analyzing the similarities and differences in the arcing characteristics under different arc-generating modes (Section 3). Based on the analysis result, a dedicated arc fault feature set is constructed (Section 4), enabling the design of an interpretable arc fault detection algorithm (Section 5). The effectiveness and robustness of the proposed method are then evaluated through comprehensive validations and comparative analyses (Section 6). Finally, Section 7 concludes the paper by summarizing the main findings and contributions.

2. Experimental platform and dataset creation

Currently, several major countries and regions have established testing standards for AFDDs. For instance, Underwriters Laboratories (UL) in the United States has published UL 1699 [33], the International Electrotechnical Commission (IEC) has issued IEC 62606 [34], and the National Standardization Administration of China has released GB/T 31143 [35]. However, these standards only specify the test procedures that AFDDs must undergo; they do not provide methodologies for detecting SAFs. With reference to the tests outlined in these standards and in consideration of the complexity of real-world distribution systems, an experimental platform was established to simulate SAFs in building distribution systems. As shown in Fig. 2, the experimental platform consists of three main components: the experimental circuit, the arc-generating device, and the signal sampling module.

In the experimental circuit, the voltage is supplied from a real-world distribution system, operating at 220 V and 50 Hz. The load set includes not only the test loads mandated by the IEC 62606 standard but also non-standard loads commonly found in building distribution systems. The complete set specifically includes: a 46 Ω resistor, a 1400 W kettle, a 1600 W oven, a 2000 W induction cooker, a 1400 W microwave oven, a 200 W drill, a 300 W fluorescent lamp, a 400 W halogen lamp, a 780 W

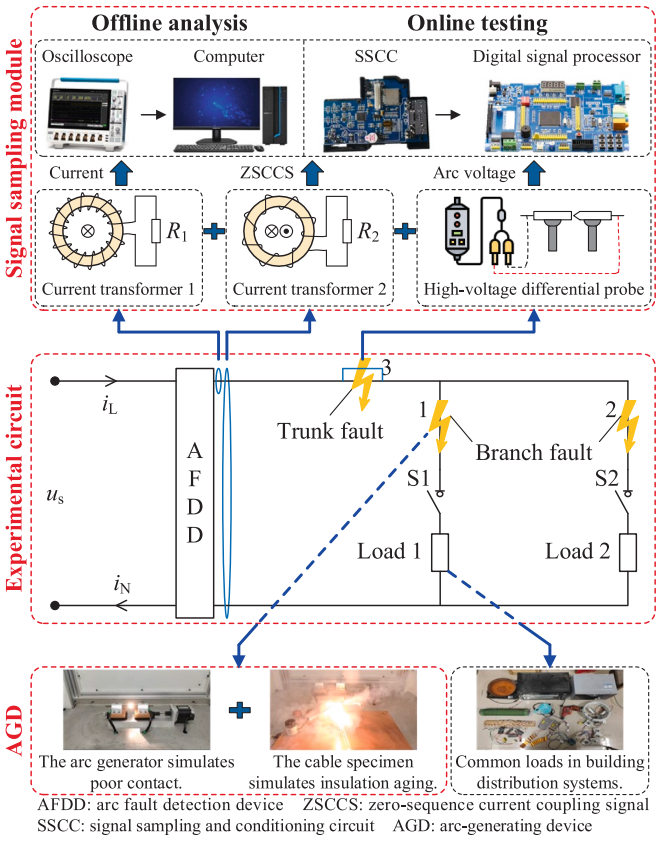


Fig. 2. Schematic diagram of the SAF experimental platform.

switching power supply, and a 1000 W dimmer with conduction angles of 30°, 60°, 90°, and 120°.

In building distribution systems, SAFs may occur either in the main circuit formed by parallel branches or in individual load branches. Based on the circuit topologies outlined in the IEC 62606 standard and those prevalent in real-world distribution systems, four representative circuit topologies were selected to simulate SAFs at different locations. These include: (A) single load circuits with a SAF at position 3 (S1 closed, S2 open); (B) combined load circuits with a SAF at position 3 (S1 and S2 closed); (C) parallel configurations with a high-power normal branch and a low-power fault branch, featuring a SAF at position 1 (S1 and S2 closed); and (D) parallel configuration with a high-power fault branch and a low-power normal branch, featuring a SAF at position 2 (S1 and S2 closed). The inclusion of diverse loads and circuit topologies ensures that the experimental platform closely replicates real-world operating conditions in building distribution systems, thereby enabling a comprehensive analysis of arc fault characteristics across various fault conditions.

The standards UL 1699, IEC 62606, and GB/T 31143 each specify two methods for simulating SAFs: one using an arc generator and the other a carbonized cable specimen. Furthermore, the vast majority of existing studies employ only one of these two arc-generating modes, without adopting alternative methods. As these two modes simulate SAFs caused by poor electrical connections and insulation degradation, respectively, they collectively represent the primary causes of SAFs in building distribution systems [36]. To ensure comprehensive coverage of these primary failure mechanisms, the arc-generating device in the experimental platform incorporates both an arc generator and a carbonized cable specimen. The arc generator incorporates four primary components: a movable electrode, a fixed electrode, a precision ball screw mechanism, and dielectric support structures. Manual rotation of the ball screw induces electrode separation, establishing a conductive plasma channel through air ionization. For cable specimen preparation,

a two-core BVVB cable—among the most common types in building distribution systems—was selected. This cable features oxygen-free pure copper conductors and new polyvinyl chloride insulation, with a cross-sectional area of 1.5 mm^2 per core. The overall outer diameter, sheath thickness, and insulation thickness are 7.5 mm, 0.9 mm, and 0.7 mm, respectively. The cable specimen is subjected to a 30 mA short-circuit current and a minimum 7 kV open-circuit voltage for a duration of 10 s, inducing carbonization. This process creates a conductive carbonized pathway along the fractured conductor, characterized by the formation of pyrolytic carbon structures from polymer decomposition, primarily consisting of polyolefin derivatives, cyclic hydrocarbon compounds, and dispersed carbonaceous particles [36]. Subsequent arc fault simulation is achieved by integrating the pre-conditioned carbonized cable specimen into the experimental circuit.

During the arc fault characteristic analysis, an oscilloscope (MSO46) is used to collect arc voltage, current, and ZSCCS at a sampling rate of 1 MHz and a time window of 20 ms, allowing for subsequent analysis of the obtained experimental data on a computer. The arc voltage is measured using a high-voltage differential probe (THDP0200, bandwidth: 200 MHz). The current and ZSCCS are first converted into voltage signals using current transformers DL-CT08CL10 and DL-CT03C0.2T, respectively, and then measured with two identical voltage probes (TPP0250, bandwidth: 250 MHz).

This study employs the aforementioned experimental platform to simulate SAFs in real-world distribution systems by changing load types, circuit topologies, and arc-generating modes. The experimental design incorporates thirteen distinct loads, four circuit topologies, and two arc-generating modes, facilitating the simulation of 104 ($C_2^1 \cdot C_{13}^1 \cdot C_4^1$) fault conditions and 52 ($C_{13}^1 \cdot C_4^1$) normal conditions. For each fault condition, 90 data samples were collected, whereas 180 data samples were collected for each normal condition, yielding a balanced dataset comprising 9360 fault samples and 9360 normal samples. This dataset will provide fundamental data support for subsequent research on: (1) comparative analysis of arc characteristics under different arc-generating modes, (2) determination of boundary thresholds between normal and fault states, and (3) performance comparison of feature sets proposed in different methods.

3. Comparative analysis of arc characteristics under different arc-generating modes

The physical development process of AC arc faults can be divided into four stages: arc-igniting, arcing, arc-extinguishing, and zero-current. In the arc-igniting stage, the circuit transitions from a cutoff to a conductive state. In the arcing stage, the circuit remains conductive. In the arc-extinguishing stage, the circuit state transitions from conduction to insulation. In the zero-current stage, the electrode gap remains insulated as the electrode voltage decreases. In the following sections, we will conduct a stage-by-stage comparative analysis of arc characteristics under different arc-generating modes, with a particular focus on the waveform signatures exhibited by electrical parameters (such as voltage, current, and ZSCCS).

To systematically elucidate the intrinsic difference in arcing characteristics induced by distinct fault mechanisms, all comparative experiments in Subsections 3.1 to 3.4 adhere to a single-variable design principle: the circuit configuration remains strictly consistent, with only the arc-generating mode adjusted. Specifically, the experimental setup adopts the fixed parameters of the platform established in Section 2, including the circuit topology, cable type/length, grounding method, and load types—ensuring that distributed circuit parameters such as cable impedance and ground capacitance are kept constant through the tests. The only manipulated variable is the arc-generating mode, which corresponds to two typical SAF types in building distribution systems: those simulated by an arc generator (mimicking poor contact) and those induced by carbonized cables (mimicking insulation aging). This design eliminates interference from circuit configuration variations, enabling the observed differences in arcing characteristics to be exclusively attributed to the intrinsic mechanisms of different arc-generating modes.

3.1. Arc-igniting stage

Fig. 3 shows the waveforms of electrode voltage, current, and ZSCCS in the arc-igniting stage. When the voltage applied across the electrodes exceeds the gap discharge threshold, breakdown occurs in the electrode gap, causing the electrode voltage to rapidly drop from the supply voltage to the arc column voltage. Simultaneously, high-frequency pulses are observed in both the current and ZSCCS. The rapid changes in electrode voltage in this stage readily generate high-frequency pulses in the circuit. This fault characteristic propagates through the circuit and can be detected in the current, ZSCCS, and branch voltage [4].

Although larger spike amplitudes in electrical parameters are beneficial for detecting SAFs, their utility is limited due to susceptibility to circuit noise and their inability to reveal the fundamental physical differences between the two arc-generating modes: the arc generator and the carbonized cable. In contrast, during the arc-igniting stage, a positive correlation exists between the electrode voltage change rate (EVCR) and spike amplitudes in both the current waveform and the ZSCCS [4]. As a relative quantity reflecting the dynamic process of arc ignition, the EVCR is therefore introduced to characterize the difference between the two arc-generating modes, and is defined as shown in (1).

$$EVCR = \frac{|u_a - u_b|}{t_b - t_a} \tag{1}$$

where u_a and u_b denote the electrode voltages before and after discharge, respectively, t_a represents the discharge moment corresponding to u_a, and t_b represents the stabilization moment corresponding to u_b.

According to (1), the EVCR is extracted from 500 sets of electrode voltage waveforms. The statistical results of the EVCR under different arc-generating modes and currents are shown in Fig. 4. In the arc generator mode, the fluctuation ranges of the EVCR at different currents are 6–500 V/μs, 6–910 V/μs, 44–1200 V/μs, 14–7200 V/μs, and 25–4100 V/μs, respectively. In the cable specimen mode, the fluctuation ranges of the EVCR at different currents are 0–0.8 V/μs, 0–3 V/μs, 0–2 V/μs, 0–2 V/μs, and 0–10 V/μs, respectively. These results demonstrate that the EVCR in arc generator mode is significantly higher than that in the cable specimen mode. As previously mentioned, the circuit configuration remained identical across all tests; only the arc-generating mode was altered. Therefore, the observed differences in EVCR between the two arc-generating modes are unequivocally attributable to the distinct arc generation mechanisms.

In the arc generator mode, the velocity of charged particle movement is primarily governed by the electric field strength, whereas in cable specimen mode, it is influenced by multiple factors, including the electric field strength and the surface condition of the dielectric material. In addition, the discharge distance in the cable specimen mode (on the order of 1 mm) is larger than that in the arc generator mode (on the order of 0.1 mm). Therefore, the EVCR in the arc generator mode is more significant than that in the cable specimen mode.

Based on the preceding analysis, the following observations can be made. In the arc generator mode, the breakdown process between the electrodes occurs rapidly, resulting in a steeper pulse and stronger emission of high-frequency electromagnetic waves. Conversely, in the cable specimen mode, the breakdown process between the electrodes is slower, leading to a less steep pulse and weaker emission of high-frequency electromagnetic waves.

3.2. Arcing stage

The high-speed camera captures the arc morphology at a

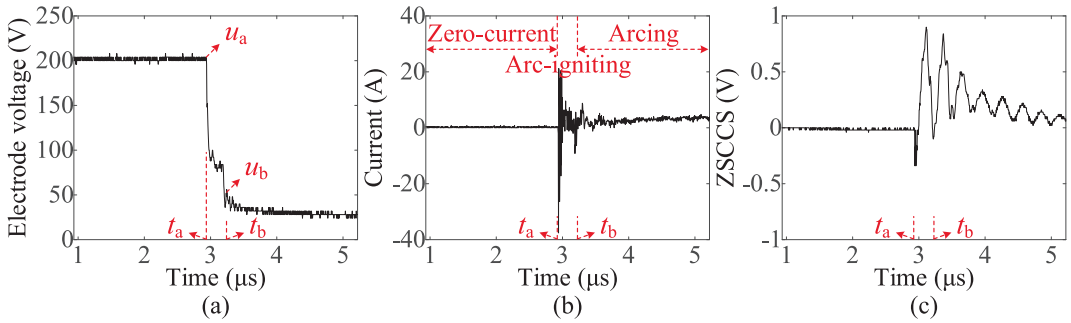


Fig. 3. Waveforms of three electrical parameters in the arc-igniting stage. (a) Electrode voltage; (b) Current; (c) ZSCCS.

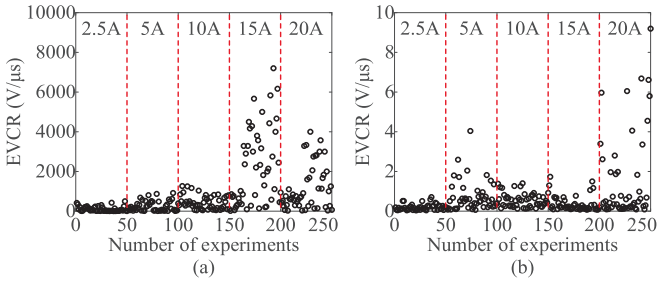


Fig. 4. EVCR under different arc-generating modes. (a) Arc generator mode; (b) Cable specimen mode.

microsecond resolution. As illustrated in Fig. 5, the pixel count within the arcing region exhibits continuous stochastic fluctuations during each arcing stage. The fluctuation patterns show significant differences between adjacent stages, revealing non-repetitive characteristics throughout the cyclic process. These pixel variations, which quantitatively reflect arc volume dynamics, provide a reliable indicator for monitoring arc resistance changes, thereby demonstrating the fundamental instability of the SAF.

Compared to the current amplitude and fluctuation level under normal states, the reduction in amplitude and increase in fluctuation under fault conditions serve as critical indicators for detecting SAFs. However, current amplitude and waveform morphology are highly susceptible to load type variations, making them unsuitable for evaluating differences in arc characteristics between distinct arc-generating modes. In contrast, mean arc voltage, mean arc resistance, and arc voltage standard deviation are less affected by such influences. Therefore, these three parameters are selected to quantitatively evaluate the arcing characteristics under two distinct arc-generating modes—arc generator and cable specimen. The computational methods for these parameters are systematically given in (2) to (4).

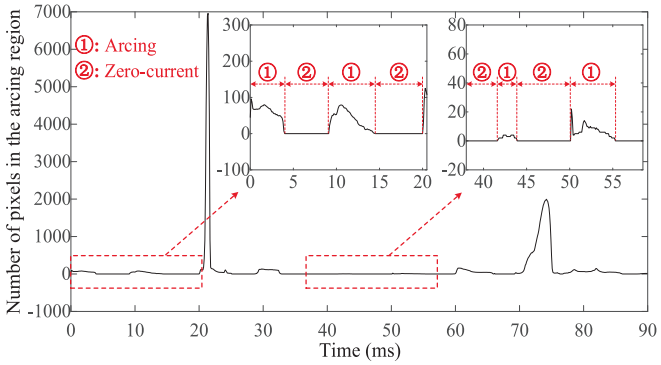


Fig. 5. The variation in the number of pixels in the arcing region.

$$\overline{u_{arc}} = \frac{1}{m}\sum_{j=1}^{m} u(j) \tag{2}$$

$$\overline{R_{arc}} = \frac{1}{m}\sum_{j=1}^{m} \frac{u(j)}{i(j)} \tag{3}$$

$$S_u = \sqrt{\frac{1}{m}\sum_{j=1}^{m} (u(j) - \overline{u_{arc}})^2} \tag{4}$$

where $u(j)$ represents the arc voltage, $i(j)$ represents the arc current, m denotes the number of signal sampling points during the arcing stage, and j indicates the index of the sampling points.

The mean arc voltage, mean arc resistance, and arc voltage standard deviation were extracted from 500 sets of arc voltage and current waveforms using (2) to (4). Fig. 6(a) presents the mean arc voltage across different arc-generating modes. The arc generator mode exhibits a lower mean arc voltage compared to the cable specimen mode, primarily due to two factors: (1) the conductive medium in the arc generator mode consists solely of air, whereas the cable specimen mode contains air and fine solid particles (carbonization products) that disrupt the arc column; (2) the conductive path length in the cable specimen mode (on the order of 1 mm) significantly exceeds that in the arc generator mode (on the order of 0.1 mm). These combined effects result in elevated arc voltage in the cable specimen mode.

Fig. 6(b) shows the mean arc resistance under different arc-generating modes, revealing two key observations. First, the mean arc resistance demonstrates an inverse relationship with arc current in both modes, attributable to three concurrent mechanisms: increased thermal energy release, enhanced charged particle density, and arc column expansion with rising current. Second, the cable specimen mode exhibits substantially higher mean arc resistance than the arc generator mode, primarily due to the presence of fine solid particles in the conductive medium and the extended conductive path length.

Fig. 6(c) displays the arc voltage standard deviation under different arc-generating modes. The cable specimen mode shows greater voltage variability than the arc generator mode, primarily due to the heterogeneous conductive medium (air and cable powder) that induces arc path variations. In contrast, the arc generator mode maintains a more stable arc path due to its homogeneous air medium, resulting in reduced voltage fluctuations.

The preceding analysis reveals key differences between the two arc-generating modes under identical circuit configurations. The arc generator mode demonstrates a higher probability of both smaller reductions in current amplitude and smaller increases in current fluctuation when compared to the cable specimen mode.

3.3. Arc-extinguishing stage

The waveforms of electrode voltage, current, and ZSCCS in the arc-extinguishing stage are presented in Fig. 7, where the electrode

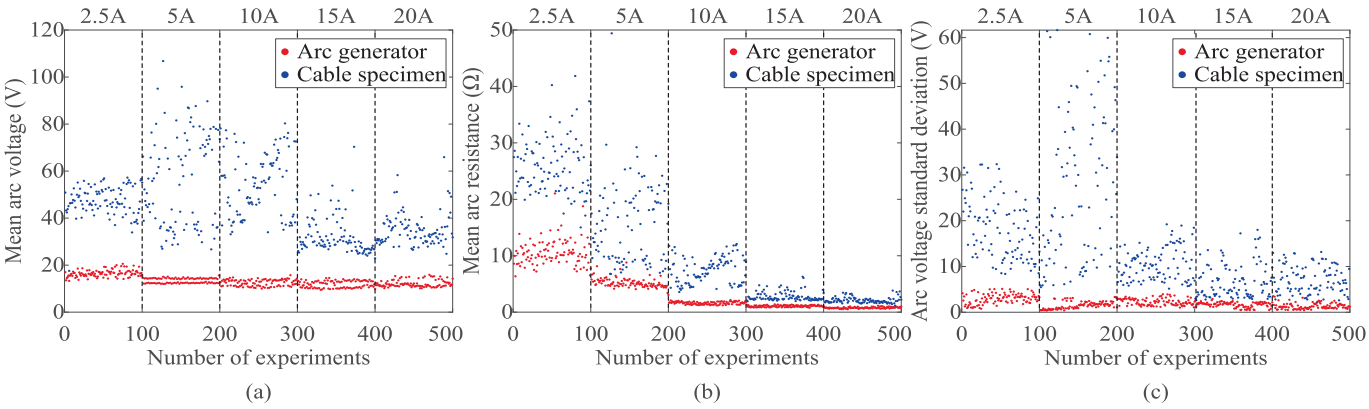


Fig. 6. Mean arc voltage, mean arc resistance, and arc voltage standard deviation under different arc-generating modes. (a) Mean arc voltage; (b) Mean arc resistance; (c) Arc voltage standard deviation.

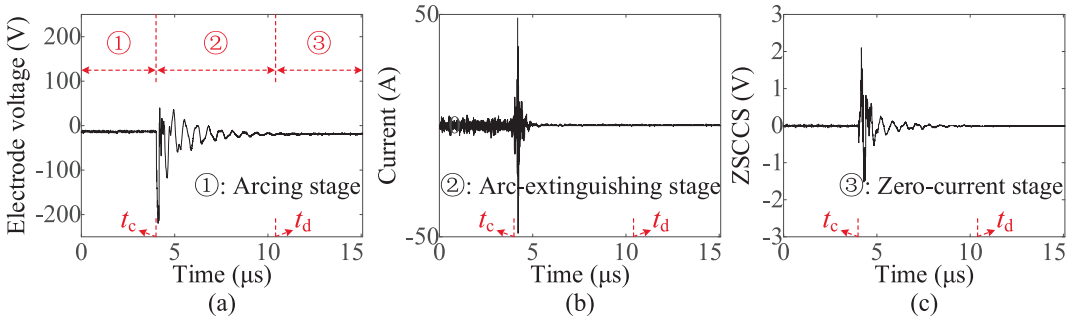


Fig. 7. Waveforms of three electrical parameters in the arc-extinguishing stage. (a) Electrode voltage; (b) Current; (c) ZSCCS.

voltage transitions from arc column voltage to supply voltage as the current progressively decays to zero. Owing to the thermal inertia of charged particle movement within the electrode gap, the current decay rate exceeds the arc resistance growth rate, leading to the occurrence of pulse spikes in branch voltage, current, and ZSCCS in this stage [4].

Similar to the arc-igniting stage, the pulse spike characteristics in electrical parameters during the arc-extinguishing stage serve as a critical basis for detecting SAFs. Both the current and ZSCCS are susceptible to load type variations, making it difficult to extract indicators that reflect differences in arc characteristics caused by different arc-generating modes. In contrast, the arc voltage remains stable and is unaffected by load types. Therefore, the number of arc voltage extinguishing spikes, the amplitude of arc voltage extinguishing spikes, and the pulse width of arc voltage extinguishing spikes are selected to quantify the differences in arc characteristics during the arc-extinguishing stage. Table 2 presents a comparative analysis of these parameters based on a statistical evaluation of 300 sets of arc voltage waveforms from both modes.

Table 2
Characteristics comparison of the arc-extinguishing stage under different arc-generating modes.

Experimental conditions		Number of arc voltage extinguishing spikes	Amplitude range of arc voltage extinguishing spikes (V)	Pulse width range of arc voltage extinguishing spikes (μs)
2.5A	Arc generator	54	(78, 284)	(5.4, 8.4)
	Cable specimen	3	(32, 63)	(0.3, 1.0)
5A	Arc generator	60	(52, 172)	(3.5, 6.2)
	Cable specimen	11	(46, 74)	(1.0, 1.9)
10A	Arc generator	35	(36, 260)	(1.0, 5.5)
	Cable specimen	10	(34, 48)	(0.7, 1.1)

The statistical analysis yields three significant findings: (1) The arc generator mode demonstrates a higher frequency of arc voltage extinguishing spikes compared to the cable specimen mode; (2) The amplitude of arc voltage extinguishing spikes is substantially greater in the arc generator mode; (3) The pulse width of arc voltage extinguishing spikes is more pronounced in the arc generator mode. These phenomena can be attributed to the enhanced stability of the arcing process in the arc generator mode, which necessitates a larger electrode voltage difference to disrupt the sustained arc. Furthermore, while the cable specimen mode exhibits non-periodic decay in arc voltage extinguishing spikes, the arc generator mode demonstrates oscillatory decay characteristics, resulting in extended pulse durations.

Based on the preceding analysis, it is evident that the transient pulse characteristics in the arc generator mode are more pronounced than those in the cable specimen mode.

3.4. Zero-current stage

Fig. 8 presents the waveforms of electrode voltage, current, and ZSCCS in the zero-current stage. When the current crosses zero, the SAF extinguishes, and the arc fault event enters the zero-current stages. In this stage, the circuit behaves as a capacitor charging process, where the electrode voltage tracks the supply voltage variations, while the current remains at zero due to the restoration of insulation between the electrode gaps. However, transient discharges may occasionally occur in the electrode gap, leading to pulse spikes in the electrical parameter waveforms.

The parameter—number of discharges in the electrode gap during the zero-current stage—is selected to quantify the differences in

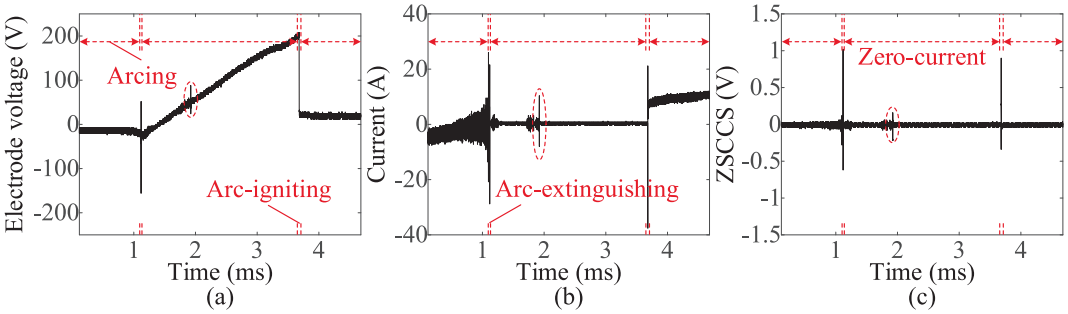


Fig. 8. Waveforms of three electrical parameters in the zero-current stage. (a) Electrode voltage; (b) Current; (c) ZSCCS.

occasional transient discharge phenomena during the zero-current stage between the two arc-generating modes. Table 3 presents the number of discharges occurring during the zero-current stage for both arc generator and cable specimen modes, based on the statistical analysis of 740 sets of arc current waveforms. The results demonstrate that the discharge frequency in the arc generator mode is significantly higher than that in the cable specimen mode. Furthermore, the discharge frequency exhibits a decreasing trend with increasing arc current.

In the arc generator mode, the conductive medium in the electrode gap is exclusively composed of air, which promotes arc reignition. Conversely, in the cable specimen mode, the conductive medium comprises both air and small solid particles, which suppress arc reignition. Additionally, the discharge distance in the cable specimen mode (on the order of 1 mm) is significantly greater than that in the arc generator mode (on the order of 0.1 mm). These factors collectively account for the higher discharge frequency observed in the arc generator mode compared to the cable specimen mode.

Compared to low-current arcing, high-current arcing is accompanied by greater energy input, a higher degree of medium ionization, and a lower arc voltage. The extended duration of arc maintenance significantly compresses the time available for the zero-current stage. Furthermore, after high-current arc extinction, the increased number of residual charged particles in the electrode gap, combined with substantial residual heat accumulated in the preheated electrodes, maintains the breakdown voltage threshold at a relatively low level during the zero-current stage. When the supply voltage rises in the subsequent half-cycle and reaches this reduced breakdown threshold, the arc rapidly refinites and transitions directly into a stable arcing stage, rather than undergoing brief unstable breakdowns. In summary, the shortened duration of the zero-current stage and the increased probability of arc reignition collectively inhibit the occurrence of multiple transient discharges, leading to a reduction in the number of instantaneous discharges as the current increases.

The zero-current duration and the standard deviation of current during the zero-current stage are selected to quantify differences in the zero-current stage and its fluctuations between the two arc-generating modes. The calculation methods for these two parameters are provided in (5) and (6), respectively.

$$T_z = t_{z1} - t_{z2} \tag{5}$$

Table 3
Relative discharge frequency in the zero-current stage under different arc-generating modes.

Experimental conditions	2.5A	5A	10A	15A	20A
Arc generator	0.72 (46/64)	0.61 (46/76)	0.36 (29/80)	0.24 (16/68)	0.29 (21/72)
Cable specimen	0.34 (23/68)	0.38 (24/64)	0.29 (30/104)	0.26 (23/88)	0.16 (9/56)

$$S_i = \sqrt{\frac{1}{n}\sum_{j=1}^{n}\left(i(j) - \frac{1}{n}\sum_{k=1}^{n} i(k)\right)^2} \tag{6}$$

where t_{z2} and t_{z1} denote the start and end moments of the zero-current stage, respectively, n represents the total number of signal sampling points within the interval $[t_{z1}, t_{z2}]$, and k indicates the index of the sampling points.

Based on (5) and (6), the zero-current duration and the standard deviation of current were extracted from 500 sets of arc current waveforms. Fig. 9(a) illustrates the differences in zero-current duration between the two arc-generating modes. A significantly shorter zero-current duration is observed in the arc generator mode compared to the cable specimen mode. This difference occurs because the higher probability of arc re-ignition in the arc generator mode under the same arc voltage conditions leads directly to an extended arcing stage and a substantially reduced zero-current duration. Fig. 9(b) demonstrates the differences in the standard deviation of current between the two arc-generating modes. The standard deviation of current in the cable specimen mode is significantly greater than that in the arc generator mode. In the arc generator mode, the arc path is stable and singular, facilitating rapid restoration of the insulating state near the current zero-crossing. In contrast, the presence of small solid particles in the cable specimen mode disrupts the conductive path, significantly delaying the restoration of the insulating state near the zero-crossing. Consequently, the cable specimen mode exhibits pronounced fluctuations during the zero-current stage, resulting in an elevated standard deviation of current.

The preceding analysis reveals that the zero-current characteristics and the fluctuations during the zero-current stage are more pronounced under the cable specimen mode than under the arc generator mode.

4. Multi-timescale arc fault feature extraction

Through the analysis of the arcing process and statistical results derived from extensive experimental data, the variations in electrical parameter waveforms caused by SAFs in building distribution systems manifest in both steady and transient states. The arcing and zero-current stages induce steady changes in electrical parameter waveforms, while the arc-igniting and arc-extinguishing stages cause transient changes. Steady characteristics are primarily defined by the volatility of electrical parameters, whereas transient characteristics are dominated by abrupt changes in these parameters. The current and ZSCCS, which are easily measurable online, excel in reflecting the steady and transient characteristics of SAFs, respectively. Therefore, this study extracts arc fault features from both current and ZSCCS.

Furthermore, in building distribution systems, there exists a category of operational conditions—such as load switching, plug insertion, and plug removal—that can also generate electrical arcs. These arcs are brief and are referred to as good arcs. If the electrical parameter waveforms of a good arc within a single cycle are compared with those of a SAF within a single cycle, it is difficult to distinguish between them. In contrast,

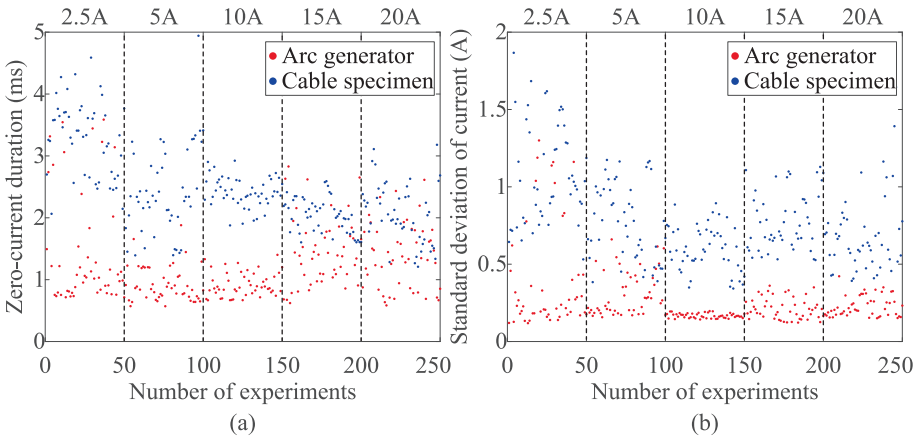


Fig. 9. Zero-current duration and standard deviation of current under different arc-generating modes. (a) Zero-current duration; (b) Standard deviation of current.

SAFs are persistent, continuously release heat, and pose significant fire hazards. The changes in current and ZSCCS caused by SAFs are reflected both within a single cycle and across multiple adjacent cycles. Therefore, the volatility of current and ZSCCS over multiple adjacent cycles serves as a critical indicator for detecting SAFs. This study comprehensively characterizes arc fault features from both single-cycle and multi-cycle perspectives.

4.1. Single-cycle arc fault feature

The wavelet transform is capable of extracting the component signals from different frequency bands of the original signal and exhibits strong processing capabilities for non-stationary signals. After the wavelet transform, the detail coefficients exclude power frequency signals and low-frequency noise, effectively mitigating the impact of various load current waveforms on arc fault feature extraction. Therefore, this study employs the mother wavelet coif1 to perform wavelet decomposition of the current signal, utilizing the first-level detail coefficients as pre-processed signals for arc fault feature extraction. In parallel, arc fault features are extracted directly from the ZSCCS without preprocessing, as this signal exhibits less influence from load types than the current signal due to magnetic field cancellation in the same current transformer.

1) Quantification of transient pulse characteristic of arc faults: Based on the analysis in Sections 3.1, 3.3, and 3.4, in addition to the sudden changes in the ZSCCS during the arc-igniting and arc-extinguishing stages, the discharge phenomenon in the zero-current stage can also cause abrupt changes in the ZSCCS. To characterize the pulse spike characteristics of the ZSCCS resulting from these abrupt changes, a pulse factor is introduced, and its calculation method is presented in (7).

$$P^{\mathrm{x}} = \frac{\max\left(|x_j|\right)}{\left(\sum_{j=1}^{N}|x_j|\right)/N} \tag{7}$$

where x_j represents the ZSCCS, and N denotes the number of signal sampling points within a single cycle.

Furthermore, the rapid switching of circuit states during the arc-igniting and arc-extinguishing stages also causes pulse spikes in the current. The detail coefficients retain the characteristics of abrupt changes in the current during these stages. The range of detail coefficients is used to characterize the difference in the abrupt changes of the current signal, and its calculation method is provided in (8).

$$R^{\mathrm{d}} = \max\left(d_j\right) - \min\left(d_j\right) \tag{8}$$

where d_j represents the first-level detail coefficients of the current signal after the wavelet transform.

The arc-igniting and arc-extinguishing moments of the SAF exhibit variability within a single cycle. Similarly, the discharge moments during the zero-current stage, when the electrode gap discharges, also show randomness. These factors collectively lead to random changes in the timing of abrupt variations in the ZSCCS. To characterize the randomness of ZSCCS change moments, the interval of pulse spikes is employed, and its calculation method is provided in (9).

$$I^{\mathrm{x}} = \left(p_{\mathrm{s}} - p_{\mathrm{f}}\right) \times \frac{1}{f} \tag{9}$$

where p_{f} represents the position of the maximum value in the sequence $\{|x_1|, |x_2|, \bullet\bullet\bullet, |x_{N/2}|\}$, p_{s} represents the position of the maximum value in the sequence $\{|x_{N/2+1}|, |x_{N/2+2}|, \bullet\bullet\bullet, |x_{N}|\}$, and f denotes the signal sampling frequency.

2) Quantification of high-frequency fluctuation of arc faults: As analyzed in Sections 3.2 and 3.4, the instability of SAFs during the arcing and zero-current stages induces high-frequency fluctuations in the ZSCCS. This, in turn, leads to an increase in its dispersibility. To characterize the dispersibility, the standard deviation is employed, and the calculation method of this metric is provided in (10).

$$S^{\mathrm{x}} = \sqrt{\frac{1}{N}\sum_{j=1}^{N}\left(x_j - \frac{1}{N}\sum_{k=1}^{N}x_k\right)^2} \tag{10}$$

Furthermore, the current exhibits steady fluctuations during the arcing and zero-current stages, increasing its high-frequency components. The energy of detail coefficients is used to characterize the high-frequency component of the current signal, and the calculation method of this metric is provided in (11).

$$E^{\mathrm{d}} = \sum_{j=1}^{N/2} d_j^2 \tag{11}$$

3) Quantification of stochastic properties of arc faults: Based on the analysis in Section 3.2, the high-frequency fluctuations contribute to the disorder of the ZSCCS. To quantify the degree of chaos, the Shannon entropy is utilized, and its calculation method is presented in (12).

$$H^{\mathrm{x}} = -\sum_{j=1}^{N} x_j^2 \log\left(x_j^2\right) \tag{12}$$

Furthermore, the instability of SAFs also leads to fluctuations in the high-frequency components of the current signal, increasing the disorder of the current signal. To characterize the degree of chaos, the Shannon entropy of detail coefficients is employed, and its calculation method is provided in (13).

$$H^{\mathrm{d}} = -\sum_{j=1}^{N/2} d_j^2 \log\left(d_j^2\right) \tag{13}$$

4.2. Multi-cycle arc fault feature

As previously discussed, the instability of SAFs is evident not only within a single cycle but also across multiple adjacent cycles, primarily reflecting the variations between these cycles. The standard deviation effectively quantifies the degree of fluctuation in a sequence over different cycles. Therefore, to characterize the instability of SAFs across adjacent cycles, the standard deviations of four arc fault features—pulse factor, Shannon entropy of ZSCCS, energy of detail coefficients, and range of detail coefficients—are utilized. Their calculation methods are presented in (14) to (17).

$$M^{\mathrm{px}} = \sqrt{\frac{1}{l}\sum_{j=1}^{l}\left(P_j^{\mathrm{x}} - \frac{1}{l}\sum_{k=1}^{l}P_k^{\mathrm{x}}\right)^2} \tag{14}$$

$$M^{\mathrm{Hx}} = \sqrt{\frac{1}{l}\sum_{j=1}^{l}\left(H_j^{\mathrm{x}} - \frac{1}{l}\sum_{k=1}^{l}H_k^{\mathrm{x}}\right)^2} \tag{15}$$

$$M^{\mathrm{Ed}} = \sqrt{\frac{1}{l}\sum_{j=1}^{l}\left(E_j^{\mathrm{d}} - \frac{1}{l}\sum_{k=1}^{l}E_k^{\mathrm{d}}\right)^2} \tag{16}$$

$$M^{\mathrm{Rd}} = \sqrt{\frac{1}{l}\sum_{j=1}^{l}\left(R_j^{\mathrm{d}} - \frac{1}{l}\sum_{k=1}^{l}R_k^{\mathrm{d}}\right)^2} \tag{17}$$

where l represents the size of the selected adjacent cycle window.

Fig. 10 illustrates the distribution characteristics of four arc fault features—standard deviation of pulse factor, standard deviation of Shannon entropy of ZSCCS, standard deviation of energy of detail coefficients, and standard deviation of range of detail coefficients—under 104 fault conditions (45 samples per condition) and 52 normal conditions (90 samples per condition). The results demonstrate that these four arc fault features exhibit high distinguishability between normal and fault states under most conditions, indicating their effectiveness in representing SAFs. Furthermore, these features exhibit complementarity, and their combination enables the identification of normal and fault states under more complex operating conditions. For instance, under operating conditions involving a dimmer lamp (experiment numbers 361–1980), a switching power supply (experiment numbers 3421–3600), and an electric drill (experiment numbers 1801–1980), the standard deviations of the pulse factor, energy of detail coefficients, and range of detail coefficients exhibited limited capability in distinguishing between normal and fault states. In contrast, the standard deviation of Shannon entropy of ZSCCS demonstrated strong discriminative ability under these conditions. Similarly, under conditions involving a halogen lamp (experiment numbers 1441–1620) and a fluorescent lamp (experiment numbers 2160–2340), the standard deviation of Shannon entropy of ZSCCS showed weak discrimination, whereas the standard deviations of the pulse factor, energy of detail coefficients, and range of detail coefficients performed effectively.

5. Decision tree and decision criterion for series arc fault detection

5.1. Arc fault decision tree model

This study extracts a total of seven single-cycle features and four multi-cycle features, forming a comprehensive feature set to characterize the SAF. After completing arc fault feature extraction, it is essential to determine appropriate thresholds for each extracted feature to effectively distinguish between normal states and fault states. This critical process involves the selection of both primary and secondary arc fault features. While filter-based feature selection methods are capable of identifying the most relevant features from a multitude of parameters associated with target variables, they often fail to account for the inherent complementarity among arc fault features, which may compromise the accuracy of classification results. In contrast, embedded feature selection methods synergistically integrate feature selection with classification learning algorithms, thereby providing robust support for the development of a reliable SAF detection model. Among various embedded feature selection methods, decision trees stand out as a widely used and effective approach.

Fig. 11 presents the construction procedure of the decision tree used

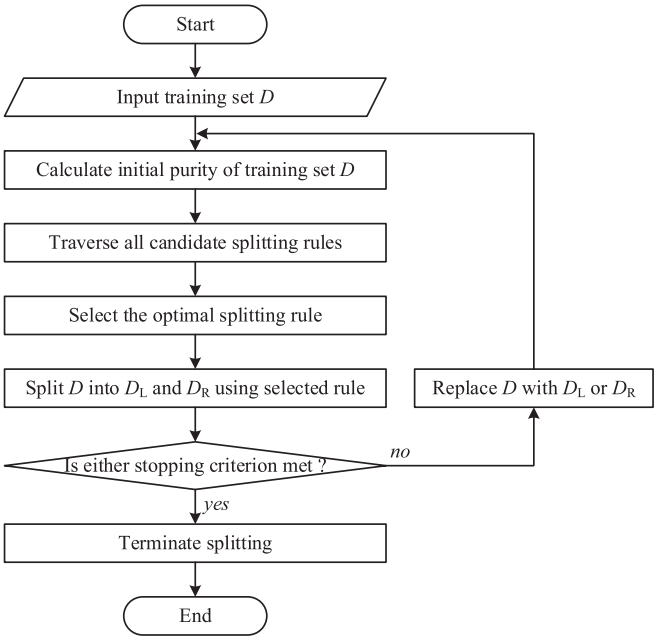


Fig. 11. Flowchart of the decision tree construction process for SAF detection.

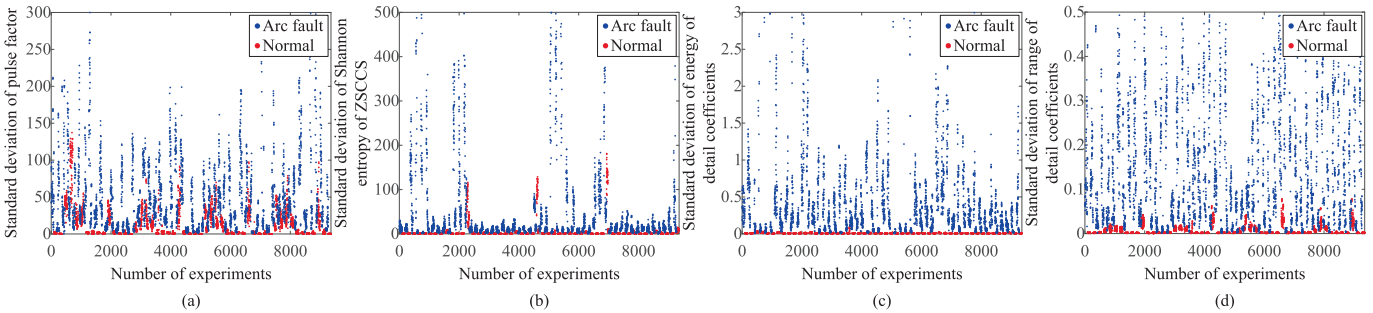


Fig. 10. Distribution characteristics of four arc fault features under different operating conditions. (a) Standard deviation of pulse factor; (b) Standard deviation of Shannon entropy of ZSCCS; (c) Standard deviation of energy of detail coefficients; (d) Standard deviation of range of detail coefficients.

for SAF detection, which comprises four key steps: (1) calculation of the initial purity, (2) traversal of candidate splitting rules, (3) selection of the optimal rule, and (4) evaluation of stopping criteria.

(1) Calculation of initial purity: The Gini coefficient of the training set D is computed using (18) to establish a purity benchmark before splitting.

$$Gi_{\mathrm{b}}(D) = 1 - \left[\left(M^{\mathrm{n}}/M\right)^2 + \left(M^{\mathrm{f}}/M\right)^2 \right] \tag{18}$$

where M denotes the total number of samples in the training set D, while M^{n} and M^{f} represent the number of samples in the training set D belonging to the normal and fault states, respectively.

(2) Traversal of candidate splitting rules: For each splitting feature (i.e., an extracted arc fault feature), its feature vector—comprising the values from all samples—is first sorted by value. The average of every two consecutive values is taken as candidate split points (i.e., decision thresholds) for that feature. Subsequently, for each candidate feature–split point pair, the training set D is partitioned into left and right subsets D_{L} and D_{R}, and the weighted Gini coefficient after the split is computed using (19). This process continues until all features and their corresponding candidate split points are evaluated.

$$Gi_{\mathrm{a}}(D) = \frac{M_{\mathrm{L}}}{M}\left[1 - \left(\frac{M_{\mathrm{L}}^{\mathrm{n}}}{M_{\mathrm{L}}}\right)^2 - \left(\frac{M_{\mathrm{L}}^{\mathrm{f}}}{M_{\mathrm{L}}}\right)^2\right] + \frac{M_{\mathrm{R}}}{M}\left[1 - \left(\frac{M_{\mathrm{R}}^{\mathrm{n}}}{M_{\mathrm{R}}}\right)^2 - \left(\frac{M_{\mathrm{R}}^{\mathrm{f}}}{M_{\mathrm{R}}}\right)^2\right] \tag{19}$$

where M_{L} and M_{R} denote the number of samples in the left and right subsets, respectively. Additionally, $M_{\mathrm{L}}^{\mathrm{n}}$ and $M_{\mathrm{L}}^{\mathrm{f}}$ represent the number of normal-state and fault-state samples in the left subset, while $M_{\mathrm{R}}^{\mathrm{n}}$ and $M_{\mathrm{R}}^{\mathrm{f}}$ correspond to those in the right subset.

(3) Selection of the optimal rule: The reduction in the Gini coefficient is calculated for each candidate rule using (20). The feature–split point pair that yields the maximum reduction is selected as the splitting rule for the current node.

$$\Delta Gi(D) = Gi_{\mathrm{b}}(D) - Gi_{\mathrm{a}}(D) \tag{20}$$

(4) Evaluation of stopping criteria: A check is performed to determine if preset stopping conditions are met—specifically, whether the tree depth exceeds ten layers or a training subset contains only one class of samples. If any condition is satisfied, the splitting process terminates; otherwise, the aforementioned steps are repeated for D_{L} and D_{R}.

A dataset matrix of size 18720 × 12 was constructed by extracting arc fault features from 9360 normal samples and 9360 fault samples and subsequently assigning labels. This matrix was then utilized to train an arc fault decision tree model using the construction procedure shown in Fig. 11. The structure of the resulting arc fault decision tree model is illustrated in Fig. 12.

The arc fault decision tree model consists of five distinct fault decision paths, each designed to identify SAFs under different conditions, as detailed below:

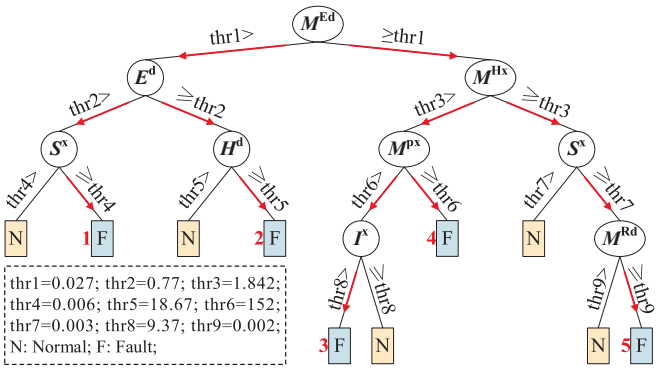


Fig. 12. Decision tree model for SAF detection.

(1) $\boldsymbol{M^{\mathrm{Ed}} \rightarrow E^{\mathrm{d}} \rightarrow S^{\mathrm{x}} \rightarrow}$ **F:** When the circuit includes interference loads such as switching power supplies or halogen lamps, the current signal contains significant harmonic components. In such cases, when SAFs are generated by the cable specimen mode, the energy of detail coefficients (E^{d}) and the standard deviation of energy of detail coefficients (M^{Ed}) exhibit minimal changes. However, the standard deviation of ZSCCS (S^{x}) shows noticeable variations. This decision path is primarily designed to detect SAFs under these operating conditions.

(2) $\boldsymbol{M^{\mathrm{Ed}} \rightarrow E^{\mathrm{d}} \rightarrow H^{\mathrm{d}} \rightarrow}$ **F:** For circuits with multiple load branches, when SAFs are generated by the cable specimen mode, the Shannon entropy of detail coefficients (H^{d}) exhibits significant changes. This decision path is mainly used to detect SAFs in such conditions. Under most normal operating conditions, the standard deviation of energy of detail coefficients (M^{Ed}) and the Shannon entropy of detail coefficients (H^{d}) remain relatively low, allowing the adjacent decision path ($M^{\mathrm{Ed}} \rightarrow E^{\mathrm{d}} \rightarrow H^{\mathrm{d}} \rightarrow$ N) to correctly classify them as normal.

(3) $\boldsymbol{M^{\mathrm{Ed}} \rightarrow M^{\mathrm{Hx}} \rightarrow M^{\mathrm{px}} \rightarrow I^{\mathrm{x}} \rightarrow}$ **F:** In circuits with arc-like loads such as dimmers, transient pulse characteristics (such as pulse spikes) appear in the current and ZSCCS, with intervals between adjacent pulse spikes being integer multiples of 10 ms. When SAFs are generated by the cable specimen mode, the standard deviation of pulse factor (M^{px}) and the standard deviation of Shannon entropy of ZSCCS (M^{Hx}) show minimal changes, while the interval time (I^{x}) varies. This decision path is specifically designed to detect SAFs under these conditions.

(4) $\boldsymbol{M^{\mathrm{Ed}} \rightarrow M^{\mathrm{Hx}} \rightarrow M^{\mathrm{px}} \rightarrow}$ **F:** In circuits with resistive loads or multiple parallel load branches (with the fault located in the main circuit), when SAFs are generated by the arc generator mode, the standard deviation of pulse factor (M^{px}) exhibits significant changes. This decision path is primarily used to detect SAFs in such operating conditions.

(5) $\boldsymbol{M^{\mathrm{Ed}} \rightarrow M^{\mathrm{Hx}} \rightarrow S^{\mathrm{x}} \rightarrow M^{\mathrm{Rd}} \rightarrow}$ **F:** Under most fault conditions, the standard deviation of energy of detail coefficients (M^{Ed}), the standard deviation of Shannon entropy of detail coefficients (M^{Hx}), the standard deviation of ZSCCS (S^{x}), and the standard deviation of range of detail coefficients (M^{Rd}) increase. This decision path effectively identifies the fault state of the circuit.

This study compares decision trees with six other common classification models in terms of accuracy, model size, computing time, interpretability, and training time, as summarized in Table 4. Notably, all timing benchmarks (computation and training) were conducted on a unified hardware configuration: a PC with AMD Ryzen 7 4800U (1.80 GHz) CPU and 16 GB DDR4 memory.

As shown in Table 4, although classification models based on support vector machines and k-nearest neighbor classifiers achieve slightly higher accuracy than that of the decision tree model, the difference is negligible. However, models trained with these two classifiers occupy greater storage resources, exhibit lower interpretability, and require longer training times. Additionally, the k-nearest neighbor classifier-based model requires significantly more computing time than the decision tree-based model, which is a critical drawback for meeting the real-time requirements of SAF detection. Compared to models trained with discriminant analysis, logistic regression classifiers, naive Bayes classifiers, and neural network classifiers, the decision tree-based model demonstrates superior accuracy, reduced model size, and faster training time. Therefore, the decision tree classifier is the most suitable choice for the arc fault feature dataset constructed in this study.

5.2. Additional arc fault decision criteria

The eleven arc fault features introduced in Section 4 are input into the trained arc fault decision tree model shown in Fig. 12. If the model identifies a SAF (flag = 'F'), the detection algorithm should not

Table 4
Comparison of different classifiers.

Name of classifier	Accuracy (%)	Size of the trained model (kB)	Computing time (ms)	Interpretability	Training time (s)
Decision trees	**99.7**	**7**	**1.842**	**easy**	**0.907**
Discriminant analysis	98.7	13	2.666	easy	0.940
Logistic regression classifiers	97.3	12	1.716	easy	1.360
Naïve bayes classifiers	96.2	9	3.773	easy	1.060
Support vector machines	99.9	14	1.809	hard	29.98
Nearest neighbor classifiers	99.8	2048	7.798	hard	3.230
Neural network classifiers	99.4	8	1.757	hard	1.929

immediately conclude that a genuine SAF has occurred. In real-world distribution systems, arc-like conditions—such as load switching, plug insertion, and plug removal—can generate waveform signatures similar to SAFs, potentially causing the arc fault decision tree model to misclassify them as SAFs. To mitigate the risk of misoperation caused by these arc-like conditions, additional arc fault decision criteria are applied after the arc fault decision tree model. Unlike transient arc-like conditions, SAFs exhibit persistent characteristics. Therefore, if more than *thr12* out of *thr11* consecutive time windows are flagged as suspected SAFs, the circuit is determined to have experienced a genuine SAF.

According to the IEC 62606 standard, an AFDD must interrupt the fault circuit within 1000 ms, 500 ms, 250 ms, 150 ms, and 120 ms for arcing currents below 2.5A, 5A, 10A, 16A, and 32A. Consequently, the values of parameters *thr11* and *thr12* are determined based on the root mean square of the current signal (I_{RMS}).

In this study, the signal sampling time window is set to 20 ms. In line with the maximum detection time allowed by the IEC 62606 standard, the parameter *thr11* takes predefined values of 50, 25, 12, 8, and 6 when I_{RMS} equals 2.5A, 5A, 10A, 16A, and 32A, respectively. For any I_{RMS} between these discrete points, *thr11* is determined by linear interpolation, with the specific calculation given in (21). In real-world distribution systems, most loads complete their startup within a single time window. However, certain loads, such as fluorescent lamps, require multiple time windows to start. Notably, fluorescent lamps operate at low currents, allowing for a longer detection time. In this study, the value of *thr12* is selected as half of the value of *thr11*.

$$thr11 = \begin{cases} 50, & I_{RMS} < 2.5A \\ \lfloor 75 - 10I_{RMS} \rfloor + 1, & 2.5 \leqslant I_{RMS} < 5A \\ \lfloor 37 - 2.4I_{RMS} \rfloor + 1, & 5 \leqslant I_{RMS} < 10A \\ \lfloor 21 - 0.8I_{RMS} \rfloor + 1, & 10 \leqslant I_{RMS} < 16A \\ \lfloor 10 - 0.125I_{RMS} \rfloor + 1, & 16 \leqslant I_{RMS} < 32A \\ 6, & I_{RMS} \geqslant 32A \end{cases} \tag{21}$$

Fig. 13 illustrates the complete flowchart of the SAF detection algorithm, which consists of three stages: (1) feature extraction from the current and ZSCCS, (2) classification by the arc fault decision tree model, and (3) final decision-making based on persistence characteristics of arc faults. The algorithm only confirms a genuine SAF if both the decision tree model detects a suspected SAF (flag = 'F') and the decision criteria are satisfied. This two-stage strategy is designed to enhance detection reliability by reducing false alarms caused by transient arc-like events.

6. Experimental validation and analysis

6.1. Offline validation

To evaluate the accuracy of the proposed SAF detection method, offline validation was conducted, consisting of closed-set tests and open-set tests. The closed-set sets were performed under operating conditions included in the training process, while the open-set tests involved

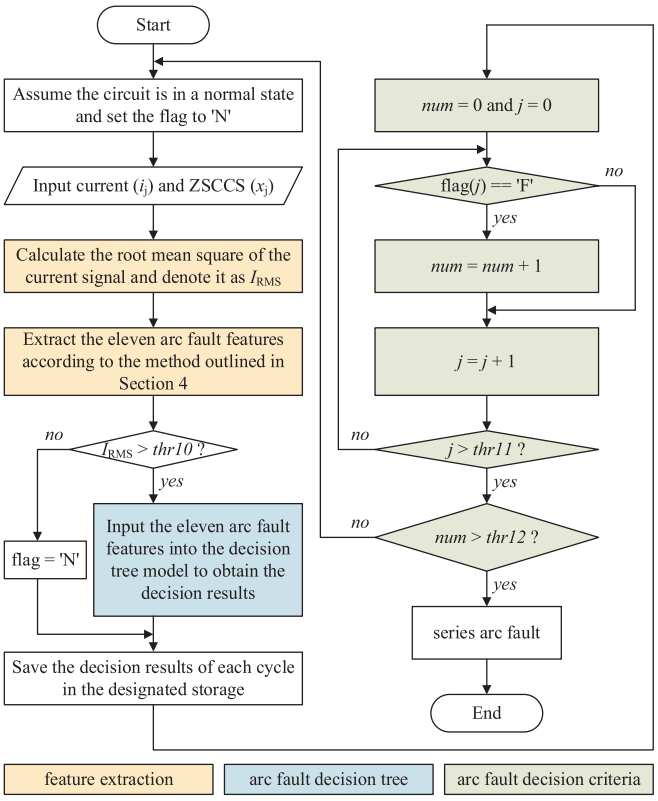


Fig. 13. Complete flowchart of the SAF detection algorithm.

conditions excluded from training but commonly found in building distribution systems or required by the IEC 62606 standard. The closed-set tests covered 104 fault conditions and 52 normal conditions, as detailed in Section 2, with 45 samples per fault condition and 90 samples per normal condition. This resulted in a total of 4680 fault samples and 4680 normal samples for closed-set testing. The open-set tests, as specified in Table 5, comprised 16 normal conditions and 16 fault conditions, with 200 samples per condition. Accordingly, the open-set tests utilized 3200 normal samples and 3200 fault samples.

Arc fault features were extracted from the aforementioned normal and fault samples following the method described in Section 4, and classification was performed using the arc fault decision tree outlined in Section 5. Fig. 14 presents the detection results of the proposed method for both closed-set and open-set testing in the form of confusion matrices. The results demonstrate that the proposed method achieves an accuracy of 99.20 % (9285/9360), with a false alarm rate of 1.07 % (50/4680) and a missed detection rate of 0.53 % (25/4680) in closed-set testing; for open-set testing, it achieves an accuracy of 99.97 % (6398/6400), with a false alarm rate of 0 % (0/3200) and a missed detection rate of 0.06 % (2/3200).

Table 5
Details of operating conditions for open-set testing.

Load Type	Vacuum cleaner	Air compressor	EMI filter 1	EMI filter 2	Line impedance	Resistance (3A, 6A, 13A, 20A, 32A)
Circuit Topology	A, B, C, D	A, B, C, D	A	A	A	A
Arc-generating Mode	Arc generator	Arc generator	Arc generator	Arc generator	Arc generator	Cable specimen

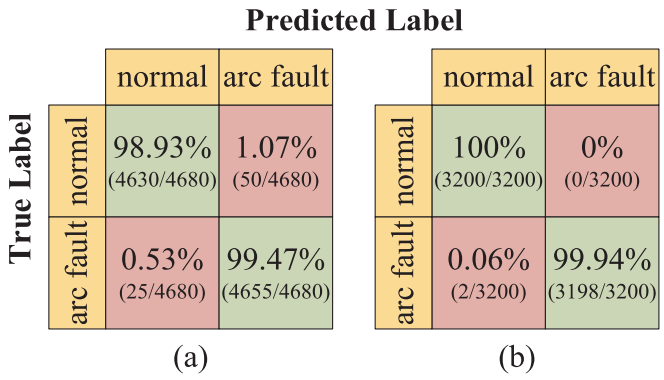


Fig. 14. Offline validation results of the proposed method. (a) Closed-set test; (b) Open-set test.

6.2. Online verification

To evaluate the real-time performance and assess the engineering application potential of the proposed SAF detection method, an AFDD prototype was developed, as shown in Fig. 15. The prototype primarily consists of four functional modules: a power supply unit, a digital signal processing (DSP) minimal unit (using the TMS320C6748 industrial microprocessor), a signal sampling unit, and a digital trip unit. The power supply unit provides regulated power to the DSP minimal unit, signal sampling unit, and digital trip unit to ensure proper operation. The signal sampling unit converts and conditions both current and ZSCCS into voltage signals suitable for analog-to-digital conversion. The DSP minimal unit performs real-time acquisition (at 1 MHz) and processes both current signals and the ZSCCS to monitor for SAFs. When a SAF is detected, the DSP minimal unit triggers the digital trip unit by issuing a trip command. Subsequently, the digital trip unit actuates the mechanical release mechanism in the miniature circuit breaker to isolate the faulty circuit.

The arc fault detection algorithm shown in Fig. 13 was ported to the prototype for online verification. The processing pipeline (feature extraction and decision-making) consistently completes in 18.577 ms for each 20-ms data window, ensuring real-time operation with a 1.423-ms temporal margin. This guarantees the complete processing of current-cycle data before the next-cycle acquisition, eliminating inter-cycle data contention. Therefore, the proposed method enables real-time arc fault monitoring.

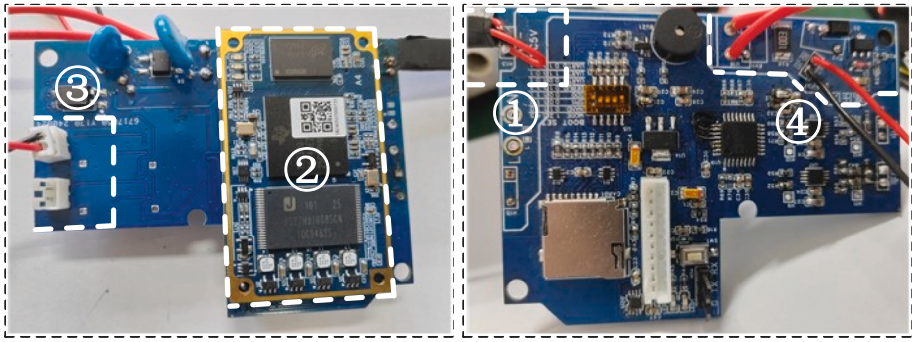
SAF experiments were conducted using the prototype under the 104 fault conditions detailed in Section 2 (closed-set test scenarios) and the 16 fault conditions listed in Table 5 (open-set test scenarios). Each condition was tested in triplicate, with the time required by the prototype to trip the fault circuit recorded for each trial; the results are presented in Fig. 16. In the closed-set test scenarios, the prototype successfully detected all SAFs, with the detection time falling within the maximum allowed by the IEC 62606 standard in 310 out of 312 tests. Similarly, in the open-set test scenarios, all SAFs were successfully detected, and the detection time complied with the standard in 44 out of 48 tests. Overall, the proposed method achieved a 98.33 % (354/360) compliance rate in terms of real-time performance for SAF detection.

6.3. Anti-interference validation

To evaluate the anti-interference performance of the proposed method under arc-like conditions, such as load switching, plug insertion/withdrawal, and transient fluctuations in building distribution systems, tests were conducted using two prototypes. The hardware configurations of the two prototypes are largely identical, with the key distinction lying in the software implementation—specifically, whether the proposed arc fault decision criterion is incorporated. The testing comprises two parts: a standard scenario anti-interference test and a generalized scenario anti-interference test.

(1) Standard Scenario Anti-interference Test: The test was conducted by connecting two prototypes in series to a circuit with the inhibition loads specified in the IEC 62606 standard. Each of the seven loads was repeatedly switched on and off. Table 6 summarizes the number of false alarms recorded by the prototypes under the frequent load switching, both with and without the proposed arc fault decision criterion enabled. The results indicate that without the criterion, 12 false alarms occurred in 70 switching operations, whereas with the criterion enabled, no false alarms were observed under the same conditions.

(2) Generalized Scenario Anti-interference Test: The two prototypes were connected in series to a real-world distribution system for a 24-hour continuous test. The system included a variety of loads, such as a television, computer, printer, microwave, air purifier, dimmable lamp, fluorescent lamp, halogen lamp, vacuum cleaner, electric heater, hair dryer, and hand drill. Throughout the test, these loads were switched on



①: Power supply unit; ②: DSP minimal unit; ③: Signal sampling unit; ④: Digital trip unit

Fig. 15. Circuit implementation of the AFDD prototype.

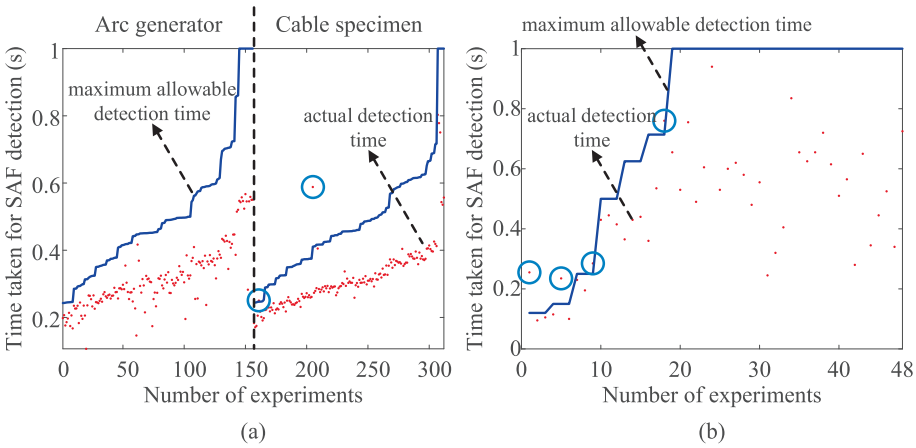


Fig. 16. Time consumed for SAF detection and the maximum allowable detection time. (a) Close-set test scenarios; (b) Open-set test scenarios.

Table 6
Comparison of false alarms with and without the arc fault decision criterion during frequent switching of inhibition loads.

Load Name	Vacuum cleaner	Switching power supply	Air compressor	Dimmer	Fluorescent lamp	Halogen lamp	Drill
Criterion Disabled	0/10	0/10	0/10	0/10	10/10	2/10	0/10
Criterion Enabled	0/10	0/10	0/10	0/10	0/10	0/10	0/10

and off at random intervals, with some operating in different modes. Results demonstrated that 240 false alarms were recorded when the proposed arc fault decision criterion was disabled, while no false alarms occurred when the criterion was enabled.

6.4. Comparative analysis and discussion

(1) Ablation Study: An ablation study was conducted to evaluate the contribution of the proposed dual-signal multi-timescale feature extraction framework. Several distinct feature sets were compared, including single-cycle features, multi-cycle features, and their hybrid combinations. For each set, an arc fault decision tree was trained and evaluated using five-fold cross-validation. As summarized in Table 7, the decision tree utilizing the hybrid feature set achieved the highest accuracy, together with the lowest false alarm and missed detection rates.

(2) Comparative Analysis with Existing Methods: Recent studies have introduced various advanced methods for SAF detection, such as those based on novel detection signals [11,26], machine learning classifiers [18,22], and hybrid time–frequency feature fusion [21]. To provide a fair and thorough evaluation of the proposed method, five of these representative techniques were selected for comparison. The assessment was based on key performance indicators, including detection accuracy, false alarm rate, missed detection rate, computational complexity of feature extraction, number of parameters in the state classification stage, and suitability for implementation on microprocessors. The comparative study followed a standardized procedure: feature sets were first extracted from a unified dataset (containing 9360 fault and 9360 normal samples) using the methodologies from [11,18,21,22], and [26]. Subsequently, classifiers were trained in accordance with the training protocols specified in each referenced study. Finally, all models were evaluated on the independent test set described in Section 6.1, which comprises 4680 fault and 4680 normal samples. The comprehensive results of this comparison are presented in Table 8.

In terms of detection performance, the proposed method outperforms the five benchmark methods, achieving higher accuracy, a lower false alarm rate, and a lower missed detection rate. This enhancement is attributed to the proposed dual-signal multi-timescale feature extraction framework, which is designed to explicitly capture the transient pulses, high-frequency fluctuations, and stochastic nature of arc faults, thereby significantly improving the quantification of fault characteristics.

Regarding computational efficiency, the proposed method exhibits lower computational complexity in the feature extraction stage than the methods introduced in [11,18,22], and [26]. Furthermore, it requires fewer parameters in the state classification stage compared to the methods presented in [18] and [22]. Although reference [11] discusses the potential for embedding their method into a microprocessor, systematic experiments are not conducted to validate the real-time performance. In addition, references [18,21,22], and [26] do not address the feasibility of deploying their methods in embedded systems for online detection. In contrast, extensive validation on a self-developed prototype has verified the real-time performance of the proposed method under both closed-set and open-set test scenarios, thereby demonstrating its practical potential.

(3) Scope of Validation and Robustness: Table 1 shows that existing SAF detection methods have been validated under a maximum of 48

Table 7
Results of the ablation study.

	With features P^x, I^x, S^x, H^x	With features P^x, I^x, S^x, H^x, M^{px}, M^{Hx}	With features R^d, E^d, H^d	With features R^d, E^d, H^d, M^{Ed}, M^{Rd}	With features P^x, I^x, S^x, H^x, R^d, E^d, H^d	With features M^{px}, M^{Hx}, M^{Ed}, M^{Rd}	**With features P^x, I^x, S^x, H^x, R^d, E^d, H^d, M^{px}, M^{Hx}, M^{Ed}, M^{Rd}**
Accuracy (%)	92.19	98.36	94.03	99.52	95.81	99.00	**99.69**
False alarm rate (%)	7.16	1.45	5.06	0.49	0.92	0.57	**0.31**
Missed detection rate (%)	8.46	1.83	6.87	0.46	7.47	1.43	**0.31**

Table 8
Comparative results of different arc fault detection methods.

Methods	Accuracy (%)	False alarm rate (%)	Missed detection rate (%)	Computational complexity of feature extraction	Number of parameters in the state classification stage	Suitability for implementation on microprocessors
[11]	90.09	15.04	4.79	O(NlogN)	3	yes
[18]	84.54	2.36	28.56	O(NlogN)	≈2000	no
[21]	89.17	5.07	16.59	O(N)	Not specified	no
[22]	90.18	6.75	12.90	$O(N^2)$	≈2500	no
[26]	84.76	4.84	25.64	O(NlogN)	2	no
Proposed method	**99.20**	**1.07**	**0.53**	**O(N)**	**12**	**yes**

operating conditions. In comparison, the proposed method was evaluated under 104 fault conditions as described in Section 2, and its effectiveness was further verified in open-set test scenarios—which reflect real-world distribution system environments not included in training—through offline, online, and anti-interference experiments. This represents an extension of operational coverage by 117 %, as quantified by (22).

$$U_p = \frac{N_c - N_o}{N_o} \times 100\% \tag{22}$$

where N_c is the number of operating conditions covered by the proposed method, and N_o is the number covered by existing methods.

7. Conclusion

This paper proposes a SAF detection method for building distribution systems that integrates a novel dual-signal multi-timescale feature extraction framework, an arc fault decision tree, and decision criteria specifically designed to mitigate arc-like interference. A comparative analysis of the arcing process under different arc-generating modes reveals that SAFs generate distinct transient and steady-state characteristics in both current and ZSCCS. Transient characteristics manifest as abrupt changes during the arc-igniting and arc-extinguishing stages, whereas steady-state characteristics exhibit fluctuations during the arcing and zero-current stages. Under test conditions encompassing thirteen loads, four circuit topologies, and two arc-generating modes, the proposed dual-signal multi-timescale feature extraction framework effectively quantifies the transient pulses, high-frequency fluctuations, and stochastic nature of arc faults. The proposed method achieves the highest detection accuracy of 99.20 % (9285/9360), outperforming five other representative methods. Online and anti-interference tests conducted on a self-developed prototype demonstrate the strong real-time performance and robustness of the designed two-stage decision logic. Among 360 online tests—covering conditions both included and not included in training but present in real-world distribution systems—the detection time met requirements in 354 tests, with zero false alarms observed under arc-like conditions such as load switching.

In this study, increasing the stringency of arc fault decision criteria reduces false alarms but prolongs detection time. Future research will, therefore, focus on resolving this compromise by optimizing the state classification and threshold selection within the current decision logic to achieve a dynamic balance between ultra-low false alarm rates and minimized detection time. Concurrently, efforts will be dedicated to enhancing the method's resilience in complex real-world distribution systems by developing more precise and robust mathematical representations of arcing characteristics based on the established feature extraction framework, thereby improving performance in high-interference environments and circuits with parallel branches.

CRediT authorship contribution statement

Zhipeng He: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Hu Zhao:** Writing – review & editing, Resources, Methodology. **Da Li:** Writing – review & editing. **Weilin Li:** Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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